

POAMS: LECTURE 9
UNIFORM MOTION: RELATIVISTIC LINEAR MOMENTUM,
MASS AND ENERGY

This lecture concerns the study of some relativistic dynamics, in particular, linear momentum, force, mass and energy. As a consequence, we shall obtain Einstein's fundamental result concerning the equivalence of mass and energy. This lecture is again based on some of 'Light Speed...' Chapter 4.

1. Some relativistic dynamics

We begin by briefly studying some of the basic concepts of *classical dynamics*. These ideas will be used in later sections and lectures.

Let O be any observer and suppose that P is a moving material object which follows a path in space relative to O . At any instant of time, t , relative to O , the *position vector*, $\mathbf{r}(t)$, of P is given by

$$\mathbf{r}(t) = (x(t), y(t), z(t))$$

where x , y and z are Cartesian coordinates for O 's frame. The (observational) **velocity**, $\mathbf{v}(t)$, of P relative to O gives the rate of change of P 's position, so that

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

a vector quantity. [See definitions (6) and (7) of Lecture 8.] Suppose now that P has mass $m(t)$. In classical dynamics, the (**linear**) **momentum**, $\mathbf{p}(t)$, of P is defined by

$$\mathbf{p}(t) = m(t)\mathbf{v}(t) \quad (1)$$

The **force**, $\mathbf{f}(t)$, exerted on the moving object P is then defined as the rate of change of momentum, so that

$$\mathbf{f}(t) = \frac{d\mathbf{p}}{dt} \quad (2)$$

Note that both $\mathbf{p}$ and $\mathbf{f}$ are vector quantities, with $\mathbf{p}$ parallel to $\mathbf{v}$. Definition (2) is usually known as *Newton's second law*. If the mass, m , of P is *constant*, then it follows by definitions (1) and (2) that

$$\mathbf{f}(t) = \frac{d(m\mathbf{v})}{dt} = m \frac{d\mathbf{v}}{dt} = m\mathbf{a}(t) \quad (2a)$$

where $\mathbf{a}(t)$ is P 's (observational) **acceleration**. [See *Lecture 8, definition (8)*.] In this case, if no force is exerted on P , then it follows that $\mathbf{a}(t) = \mathbf{0}$, so that $\mathbf{v}(t)$ is constant and the moving object P moves in a straight line relative to O . This is essentially *Newton's First Law*.

Newton's third law, or the law of action and reaction, states that if two material objects act on each other, then the force exerted on the second is equal in magnitude and opposite in direction to the force exerted by the second on the first. It follows that the total of all the forces exerted in a closed system of objects is net zero, so that by definition (2), we have,

Conservation of (linear) momentum: *in any system of objects not subjected to external forces, the total (linear) momentum is constant.*

Once again, the classical definitions, this time of momentum and force, are not satisfactory since they are observer-dependent. However, recalling the definition of four-velocity given in the last lecture, it should be clear the following relativistic definitions, analogous to the classical definitions, are the natural ones to make.

Recall that the *position vector*, $\mathbf{R}(\tau)$, of any *event* on the world-line of a material object, P , in Minkowski space-time, at any instant of *proper time*, τ , is defined by

$$\mathbf{R}(\tau) = (t(\tau), x(\tau), y(\tau), z(\tau))$$

Then the *four-velocity*, $\mathbf{U}(\tau)$, of P at any instant of proper time is given by

$$\mathbf{U}(\tau) = \frac{d\mathbf{R}}{d\tau}$$

[Lecture 8, definitions (9) and (10).] Let P have **rest mass** $m_0(\tau)$, defined as the mass of P as recorded in P 's own *rest frame*. This is independent of the observer. Then the **four-momentum**, $\mathbf{P}(\tau)$, of P is defined by

$$\mathbf{P}(\tau) = m_0(\tau)\mathbf{U}(\tau) \quad (3)$$

and the **four-force**, $\mathbf{F}(\tau)$, exerted on P is defined by

$$\mathbf{F}(\tau) = \frac{d\mathbf{P}}{d\tau} \quad (4)$$

Compare the relativistic definitions (3) and (4), with the classical definitions (1) and (2). These relativistic definitions are clearly independent of the observer.

We shall now consider only the case in which the rest mass, m_0 , of P is *constant*. It follows directly from definition (3) and equation (14) of Lecture 8, that is,

$$\mathbf{U}(\tau) = \gamma(t)(1, \mathbf{v}(t))$$

that,

$$\mathbf{P}(\tau) = m_0\gamma(t)(1, \mathbf{v}(t)) \quad (5)$$

where $\mathbf{v}(t)$ is the classical (observer-dependent) velocity of P and $\gamma(t)$ is given by

$$\gamma(t) = (1 - v^2(t)/c^2)^{-1/2} \quad (6)$$

[as in equation (13a), Lecture 8] where $v(t) = \|\mathbf{v}(t)\|$ is the *speed* of P relative to O .

Relative to this observer O , the **relativistic mass**, $m(t)$, of P is defined by the coefficient occurring in equation (5), that is,

$$m(t) = m_0\gamma(t) = m_0(1 - v^2(t)/c^2)^{-1/2} \quad (7)$$

Note that even in the case we are considering, in which the rest mass, m_0 , of P is constant, the relativistic mass is a function of the observational time, t . Equation (5) then reads

$$\mathbf{P}(\tau) = (m(t), m(t)\mathbf{v}(t)) \quad (8)$$

In Minkowski space-time, it is supposed that there is a conservation of momentum law, analogous to conservation of (linear) momentum in classical dynamics:

Conservation of four-momentum: *In any system of material objects, not subjected to external forces, the total four-momentum remains constant.*

It is important to realise here that this conservation principle cannot be established simply using the postulates of Special Relativity, or our alternative Postulates 1 and 2. Hence, in either case, this conservation principle may be regarded as an additional postulate. Conservation of four-momentum implies, by equation (8), that relativistic mass, $m(t)$, is conserved. This implies that a material object P travelling at speed $v(t)$ relative to an inertial observer O , has an effective (observational) mass $m(t)$, given by equation (7), relative to O . Note here that $m(t) > m_0$ for $v \neq 0$, so that the *faster* P travels relative to O , the *heavier* it appears to be relative to O . Once again, it must be appreciated that this is only a *relative* effect. This is the *mass-increase-with-speed effect*, which has been experimentally verified beyond reasonable doubt. This effect was first detected by Walter Kaufmann (1871-1947) and later by Alfred Bucherer (1867-1927) in the early 1900's, when they were investigating beta ray radiation.

A number of different problems involving interacting 'particles' can be solved by applying conservation of four-momentum, exactly as in Special Relativity. These include *coalescence* and *elastic collision* problems. The technique is to equate the total four-momentum before and after interaction. A typical coalescence problem is given as Example 4.4, page 72, in '*Light Speed ...*'

2. Classical work and energy

In classical dynamics, *potential energy* is essentially the capacity to do *work*. For example, energy is required to move a body through a distance, and work is then done by the force required to move the body through that distance. In general, the **work**, W , done by a force $\mathbf{f}(x, y, z)$ in displacing a body along a curve C in three-dimensional space, from an initial point A to a final point B is defined by the vector line integral

$$W = \int_C \mathbf{f} \cdot d\mathbf{r} \quad (9)$$

where $d\mathbf{r} = (dx, dy, dz)$. Suppose that the force $\mathbf{f}(x, y, z)$ is **conservative**, so that it takes the form

$$\mathbf{f}(x, y, z) = -\left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)$$

for some function $V(x, y, z)$. It then follows from definition (9) that

$$W = -\int_A^B \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz = -\int_A^B dV = V(A) - V(B) \quad (10)$$

so that W is *independent* of the curve joining A to B . In this case it follows that the work done by the force around a *closed* curve is zero, and hence the use of the term ‘conservative’. (Conversely, if W is *independent* of the curve joining A to B , then it can be shown that $\mathbf{f}(x, y, z)$ takes the special form given above.) The function $V(x, y, z)$ defines the **potential energy** at any particular point. Equation (10) expresses the fact that the work done equates to the *loss* of potential energy when the force $\mathbf{f}(x, y, z)$ moves from point A to point B .

Now suppose that the mass, m , of the body is *constant*. It then follows by definition (9) and equation (2a) that

$$W = \int_C \mathbf{f}(t) \cdot \frac{d\mathbf{r}}{dt} dt = m \int_C \mathbf{v}(t) \cdot \mathbf{a}(t) dt \quad (11)$$

where $\mathbf{r}(t) = (x(t), y(t), z(t))$ is the equation of the curve C , $\mathbf{v}(t)$ is the velocity of the body and $\mathbf{a}(t)$ is its acceleration. Note that if, once again, $v(t) = \|\mathbf{v}(t)\|$ denotes the body’s speed, then

$$\frac{d}{dt}(v^2) = 2v \frac{dv}{dt} = \frac{d}{dt}(\mathbf{v} \cdot \mathbf{v}) = 2\mathbf{v} \cdot \frac{d\mathbf{v}}{dt} = 2\mathbf{v}(t) \cdot \mathbf{a}(t) \quad (12)$$

Hence, from equation (11),

$$W = \frac{m}{2} \int_A^B \frac{d}{dt}(v^2) dt = \frac{m}{2}(v^2(B) - v^2(A)) \quad (13)$$

independently of the curve C joining A to B .

The **kinetic energy** of a moving body with mass m and speed $v(t)$ at any instant of time is denoted and defined by

$$K(t) = \frac{1}{2} m v^2(t) \quad (14)$$

Then equation (13) expresses the fact that when work is done by moving a force $\mathbf{f}(x, y, z)$ from point A to point B , there is a *gain* in kinetic energy. In other words, as a consequence of doing work, some potential energy is converted into kinetic energy. The **total energy**, E , is defined by

$$E = K(t) + V(t) \quad (15)$$

It follows by equations (10) and (13) that, in this case, the total energy is always constant. This is the *law of conservation of energy*.

3. Relativistic mass and energy

Once again, the classical treatment of the concept of energy requires modification in Special Relativity. We now show, in our approach, how Einstein's famous formula, which relates relativistic mass and energy, essentially *follows from the time-dilation formula*.

Consider again a material object, P , travelling at an (observational) velocity $\mathbf{v}(t)$ relative to an inertial observer O . It follows by the time-dilation formula, equation (12), Lecture 8, that

$$\frac{dt}{d\tau} = (1 - v^2(t)/c^2)^{-1/2}$$

at any instant of proper time, τ , where $v(t) = \|\mathbf{v}(t)\|$ is the speed of P . Using the definition, (6), of the ‘gamma factor’, this can be written as

$$\frac{dt}{d\tau} = (1 - v^2(t)/c^2)^{-1/2} = \gamma(t) \quad (16)$$

Using the definition of $\gamma(t)$ and equation (12) it follows that

$$\frac{d\gamma}{dt} = -\frac{1}{2} \left(-\frac{2v}{c^2} \frac{dv}{dt} \right) \left(1 - \frac{v^2}{c^2} \right)^{-3/2} = \frac{\gamma^3 v}{c^2} \frac{dv}{dt} = \frac{\gamma^3}{c^2} (\mathbf{v} \cdot \mathbf{a})$$

where $\mathbf{a}(t)$ is the (observational) acceleration of P . It follows from this result and the definition of $\gamma(t)$ again, given in equation (16), that

$$\mathbf{v} \cdot \frac{d}{dt} (\gamma \mathbf{v}) = (\mathbf{v} \cdot \mathbf{v}) \frac{d\gamma}{dt} + \gamma (\mathbf{v} \cdot \mathbf{a}) = c^2 \frac{d\gamma}{dt} \left(\frac{v^2}{c^2} + \frac{1}{\gamma^2} \right) = c^2 \frac{d\gamma}{dt}$$

It follows from this second result that if P has *constant* rest mass m_0 , then

$$c^2 \frac{d(\gamma m_0)}{dt} = \mathbf{v} \cdot \frac{d}{dt} (\gamma m_0 \mathbf{v}) \Rightarrow c^2 \frac{dm}{dt} = \mathbf{v} \cdot \frac{d}{dt} (m \mathbf{v})$$

where $m(t)$ is the relativistic mass of P , defined by equation (7). Hence, if P is initially at rest at time t_0 and attains velocity $\mathbf{v}$ at time t relative to observer O , then letting $\mathbf{p}(t) = m(t)\mathbf{v}(t)$ denote the relativistic (linear) momentum of P at any time t , (see definition (1)), gives

$$c^2 \int_{t_0}^t \frac{dm}{dt} dt = \int_{t_0}^t \mathbf{v} \cdot \frac{d}{dt} (m \mathbf{v}) dt = \int_{t_0}^t \mathbf{v} \cdot \frac{d\mathbf{p}}{dt} dt$$

Using definition (2), and the definition of the observational velocity, this then gives

$$c^2 (m(t) - m_0) = \int_{t_0}^t \frac{d\mathbf{r}}{dt} \cdot \mathbf{f} dt = \int_{t_0}^t \mathbf{f} \cdot d\mathbf{r} \quad (17)$$

where $\mathbf{f}(t)$ is the force exerted on P at any time t .

It follows by definition (9) that the right-hand-side of equation (17) represents the work done in accelerating P from rest and so by equation (13), the gain in the *relativistic kinetic energy* of P must be $(m(t) - m_0)c^2$. This fact is supported by calculating this quantity by using the definitions of $\gamma(t)$ and $m(t)$ given in equation (7) as follows.

$$(m(t) - m_0)c^2 = (\gamma(t) - 1)m_0c^2 = m_0c^2((1 - v(t)^2/c^2)^{-1/2} - 1)$$

where $v(t)$ is the speed of P . Since $v^2/c^2 < 1$, we may expand $\gamma(t)$ in a convergent binomial series to obtain,

$$\begin{aligned} (m(t) - m_0)c^2 &= m_0c^2 \left(1 - \frac{1}{2} \left(-\frac{v^2}{c^2} \right) + \frac{1}{2!} \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) \left(-\frac{v^2}{c^2} \right)^2 + \dots - 1 \right) \\ \Rightarrow (m(t) - m_0)c^2 &\cong m_0c^2 \left(\frac{v^2}{2c^2} + \frac{3v^4}{8c^4} \right) = \frac{1}{2}m_0v^2 + \frac{3}{8}m_0\frac{v^4}{c^2} \end{aligned}$$

If $v(t)$ is much smaller than c , this result shows that $(m(t) - m_0)c^2$ is approximately equal to the Newtonian kinetic energy of P , given by (14). The other terms in the infinite series for $(m(t) - m_0)c^2$, of which $3m_0v^4/8c^2$ is the dominant term, represent the relativistic correction.

Once again, let an object, P , have rest mass m_0 , and relativistic mass $m(t)$ relative to an observer O . From the arguments just presented, we may define the **relativistic kinetic energy** of P as $(m(t) - m_0)c^2$. Then bearing in mind the classical definitions of potential and total energy (see definition (15)), given previously, we define the **internal energy**, E_0 , of P by

$$E_0 = m_0c^2 \quad (18)$$

and the **total energy**, $E(t)$, at any time t relative to O , by

$$E(t) = m(t)c^2 \quad (19)$$

The **relativistic kinetic energy** of P at any time t relative to O is then $E(t) - E_0$. Note that the internal energy is independent of the observer and is the relativistic analogue of the potential energy. Since c^2 is such a large factor, any object contains an enormous amount of internal energy in relation to its rest mass. Even if a particle has a minute rest mass, its internal energy is still huge.

It follows from definition (19) and definition (7) of the relativistic mass, that

$$E(t) = (1 - v^2/c^2)^{-1/2} m_0 c^2$$

This relation provides yet another argument against the interpretation of c as a speed. For if c was a speed, then the amount of energy required to accelerate an object from a speed less than c to ‘speed’ c would be theoretically infinite, so that such an acceleration is not physically possible. Rather, in keeping with our Normal Realist approach, definitions (18) and (19) show that the universal constant c^2 acts as a *conversion factor* for converting mass into energy in the same way that c acts as a conversion factor for converting observational time into observational distance. Then definitions (18) and (19) demonstrate Einstein’s equivalence of mass and energy so that when we refer to *mass*, what we are really referring to is *energy*.

It is now well known that energy is realised upon the break-up of the nucleus of any chemical element with atomic weight greater than 108, in a process known as *nuclear fission*. Experiments in nuclear fission were first performed by John Cockcroft (1897-1967) and Ernest Walton (1903-1995) in Cambridge, in 1932. It was found that the amount of energy released upon the break-up of lithium nuclei was as predicted by the formula (18). Following this pioneering work, experiments concerning uranium fission were performed by Otto Hahn (1879-1968), Lise Meitner (1878-1968) and Fritz Strassmann (1902-1980), in Berlin, in 1938. The fact that uranium fission is self-sustaining, thereby producing a nuclear chain reaction, was discovered by Enrico Fermi (1901-1954). It was the advice from Einstein, Fermi and his co-worker Leo Szilard (1918-1964), all of whom had fled to

the US from Europe before the start of the Second World War, to President Roosevelt concerning the potential destructive power of uranium fission which led to the *Manhattan Project* for the development of the atomic bomb. Einstein, a dedicated pacifist, was extremely reluctant to encourage the development of this bomb. In 1945 he was devastated to learn that such an atomic bomb had been dropped on Hiroshima.

Energy is released when two or more nuclei of elements with atomic weight of less than 108 are fused together in a process called *nuclear fusion*. Hydrogen fusion takes place in the Sun and all other stars. This fact was discovered independently by Hans Bethe (1906-2005) and Carl Friedrich von Weizsäcker in 1939. Their discovery solved the problem of the source of solar energy. A hydrogen bomb works on the basis of hydrogen fusion. The first such bomb, one thousand times more powerful than the first atomic bomb, was detonated as a test in the US Marshall Islands in 1952.

Having given an account of the Normal Realist approach to high- speed uniform motion in the last five lectures, in the next lecture I shall move on to discuss the ultimately discrete nature of physical phenomena and the beginnings of Quantum Mechanics.

Anthony D. Osborne, June 2023